

Inside JFinance

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1 Introduction

This document provides detailed information about the JFinance class library (version 0.2.0). This is the second release of this document.

1.1 What is JFinance?

JFinance is a collection of classes for developing financial applications in Java. This release of JFinance contains classes for representing business day calendars.

You can download the library from <http://www.jrefinery.com> and use it in your own projects, subject to the terms of the GNU General Public Licence (see the next section).

The code should compile and run on any implementation of the Java 2 platform (SDK 1.2.2 or later).

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Organisation:	Web Address:
The Free Software Foundation	http://www.fsf.org
The Open Source Initiative	http://www.open-source.org

Appendix A contains the full text of the GNU General Public Licence.

1.3 Latest Version

This document has been written for version 0.2.0 of JFinance. The latest version, plus any bug reports or other announcements, can be found on my web site, <http://www.jrefinery.com>. Bug reports and other announcements will also be posted on the site.

1.4 Comments and Suggestions

If you have any comments or suggestions regarding JFinance, or this document, please send e-mail to: david.gilbert@bigfoot.com.

2 Class Reference

This section contains special notes and reference information for all the classes in the JFinance class library.

Package: (com.jrefinery.*) **Description:**

`finance` Base classes;

`finance.data.reference` Reference data classes;

Additional information can be found in the Javadoc-generated HTML documentation for JFinance.

2.1 Package: com.jrefinery.finance

This package contains the base classes for JFinance.

2.1.1 Abstraction

The base class for representing all entities that are defined by the JFinance “model”.

It is important to keep in mind that **Abstraction** and all of its subclasses are concerned with representing entities *outside* the model. The representation *inside* the model depends on how the model has been implemented. JFinance has nothing to say about the implementation of a model—only the interfaces used to add, update and retrieve **Abstractions** from the model.

See Also

`AbstractionException`;

2.1.2 AbstractionException

The base class for all exceptions that can occur when handling an **Abstraction**. Typically, these relate to the internal consistency of the entity. They do not relate to the model.

See Also

`Abstraction`;

2.1.3 DuplicateEntityIdException

An exception (extends `ModelException`) that indicates that an attempt has been made to add an **Abstraction** to the model with an entity id that is already in use.

Note that if you add the entity with a zero id, the model will automatically assign a unique id.

Do not confuse this class with `DuplicateSubEntityIdException`, which is an `AbstractionException` rather than a `ModelException`.

See Also

`ModelException`;

2.1.4 DuplicateNameException

An exception that indicates that an attempt has been made to add an `Abstraction` to a model, with a name that is already in use.

See Also

`ModelException`;

2.1.5 DuplicateSubEntityIdException

An exception (extends `AbstractionException`) that indicates an attempt to add a sub-entity to an `Abstraction` with an entity id that is already in use.

See Also

`AbstractionException`

2.1.6 DuplicateSubNameException

An exception (extends `AbstractionException`) that indicates an attempt to add a sub-entity to an `Abstraction` with a name that is already in use.

See Also

`AbstractionException`

2.1.7 ModelException

The base class for exceptions generated by the “model”. Exceptions are generated to indicate that an operation failed because it would result in the model becoming inconsistent.

The other major class of exception in JFinance is represented by the base class `AbstractionException`. These exceptions are generated to indicate that an operation on an `Abstraction` (external to the model, remember) has failed.

See Also

`AbstractionException`;

2.1.8 NamedAbstraction

The base class for abstractions that have names and entity ids.

See Also

`Abstraction;`

2.1.9 SubEntityNotFoundException

An exception that indicates that a sub-entity within an `Abstraction` could not be found. For example, a specific holiday rule could not be found within a business day calendar.

See Also

`AbstractionException;`

2.2 Package: `com.jrefinery.finance.data.reference`

This package contains reference data classes. For now, this includes only business day calendars (and the holidays that define the calendars). In a later release, this package will also include other reference data items, such as currencies and business units.

2.2.1 AnnualHoliday

A rule that defines a specific date to be a holiday in each year.

Constructors There are two constructors:

```
public AnnualHoliday(String description, long entityId,
    BusinessDayCalendar owner, AnnualDateRule rule, boolean
    adjusted, boolean doubleHoliday)
```

The full constructor – builds an `AnnualHoliday` with the specified parameters. The `owner` is optional, set it to `null` if there is no owner for the holiday. If `adjusted` is `true`, then the holiday rolls forward if it falls on a weekend. If `doubleHoliday` is `true` then the holiday is considered to fall on two consecutive days (adjusted for weekends if requested).

```
public AnnualHoliday()
```

The default constructor – builds a default holiday (one that falls on January 1 of each year, without adjustment for weekends. The default itself is usually unimportant, you would call this constructor when you require an `AnnualHoliday` object, but it doesn't matter what it contains (this situation is rare).

See Also

`Holiday`;

2.2.2 BusinessDayCalendar

A representation of a business day calendar. You can use this class to represent business day calendars **outside** the model.

Constructors There are three constructors for this class:

```
public BusinessDayCalendar(String name, long entityId,
Weekend weekend, List holidays);
```

The full constructor – builds a business day calendar with the specified attributes.

```
public BusinessDayCalendar(String name, long entityId,
Weekend weekend);
```

A standard constructor – builds a business day calendar with the specified name, id, and weekend. There are initially no holiday rules defined for the calendar.

```
public BusinessDayCalendar(String name);
```

A standard constructor – builds a business day calendar with the specified name. The id defaults to zero, the weekend includes `SATURDAY` and `SUNDAY`, and there are initially no holiday rules defined for the calendar.

See Also

`Holiday`, `Weekend`;

2.2.3 BusinessDayCalendarReference

A placeholder for a `BusinessDayCalendar`. The reference contains the name and entity id, but no information about the holiday rules defined for the

`BusinessDayCalendar`. One typical use for a `BusinessDayCalendarReference` is constructing list of calendars for presentation to the users.

See Also

`BusinessDayCalendar`;

2.2.4 BusinessDayCalendarsFromModel

Defines the interface for obtaining `BusinessDayCalendar` objects from the model.

Methods The following methods are defined by the interface:

```
public BusinessDayCalendar getBusinessDayCalendar(String
name) throws ModelException;
```

Returns the business day calendar with the specified name.

```
public BusinessDayCalendar getBusinessDayCalendar(long
entityId) throws ModelException;
```

Returns the business day calendar with the specified entity id.

```
public BusinessDayCalendar getBusinessDayCalendar(long
entityId, int state) throws ModelException;
```

Returns the business day calendar with the specified entity id, from a specific state in the model.

```
public BusinessDayCalendar getBusinessDayCalendar(long
entityId, Date instant) throws ModelException;
```

Returns the business day calendar with the specified entity id, “as at” a specific instant in time.

```
public BusinessDayCalendar getBusinessDayCalendar(long
entityId, int state, Date instant) throws ModelException;
```

Returns the business day calendar with the specified entity id, from the specified state, “as at” a particular instant in time.

```
public Collection getBusinessDayCalendars();
```

Returns a collection containing all the business day calendars in the model.

`public Collection getBusinessDayCalendars(int scope);`
Returns a collection containing all the business day calendars in the model, using the specified scope.

`public Collection getBusinessDayCalendars(int scope, int state);`
Returns a collection containing all the business day calendars in the specified state of the model.

`public Collection getBusinessDayCalendars(int scope, Date instant);`
Returns a collection containing all the business day calendars “as at” the specified instant in time.

`public Collection getBusinessDayCalendars(int scope, int state, Date instant);` Returns a collection containing all the business day calendars in the specified state “as at” an instant in time.

See Also

`BusinessDayCalendarsToModel;`

2.2.5 BusinessDayCalendarsToModel

The interface that defines the behaviour of the “model” with regard to storing `BusinessDayCalendars` in the model.

Methods

The interface defines the following methods:

`public long addBusinessDayCalendar(BusinessDayCalendar calendar) throws ModelException;`
Adds the calendar to state zero in the model.

`public long addBusinessDayCalendar(BusinessDayCalendar calendar, int state) throws ModelException;`
Adds the calendar to a particular state in the model.

`public long addBusinessDayCalendar(BusinessDayCalendar calendar, Date instant) throws ModelException;`

Adds the calendar to state zero of the model, at a particular instant in time.

```
public long addBusinessDayCalendar(BusinessDayCalendar
calendar, int state, Date instant) throws ModelException;
```

Adds the calendar to a particular state of the model, as at the specified instant in time.

```
public void updateBusinessDayCalendar(BusinessDayCalendar
calendar) throws ModelException;
```

Modifies an existing calendar in the model (uses the entity id).

```
public void updateBusinessDayCalendar(BusinessDayCalendar
calendar, int state) throws ModelException;
```

Modifies an existing calendar in a specific state in the model (uses the entity id).

```
public void updateBusinessDayCalendar(BusinessDayCalendar
calendar, Date instant) throws ModelException;
```

Modifies an existing calendar as at a particular instant in time.

```
public void updateBusinessDayCalendar(BusinessDayCalendar
calendar, int state, Date instant) throws ModelException;
```

Modifies an existing calendar in a specific state of the model, as at a particular time.

```
public void deleteBusinessDayCalendar(long entityId) throws
ModelException;
```

Deletes the calendar with the specified entity id.

```
public void deleteBusinessDayCalendar(long entityId, int
state) throws ModelException;
```

Deletes the calendar with the specified entity id, from a particular state in the model.

```
public void deleteBusinessDayCalendar(long entityId, Date
instant) throws ModelException;
```

Deletes the calendar with the specified entity id, as at a particular instant in time.

```
public void deleteBusinessDayCalendar(long entityId, int
state, Date instant) throws ModelException; Deletes the
calendar with the specified entity id, from a particular state as
at an instant in time.
```

See Also

`BusinessDayCalendarsFromModel;`

2.2.6 Holiday

The base class for the holiday rules recorded in a `BusinessDayCalendar`.

A holiday can be adjusted for weekends—if the holiday falls on a day that is considered to be “the weekend”, then (if the `adjusted` flag is set) the holiday rolls forward to the next day that is not part of the weekend. Note that the `Holiday` object doesn’t know anything about weekends, it simply calls the `adjustForWeekend(...)` method of its owner.

You can also specify that a holiday is a “double” holiday. This means that the holiday is observed on two consecutive days (or the day immediately before a weekend, and the day immediately after a weekend). This is useful for specifying paired holidays such as Christmas Day and Boxing Day.

See Also

`AnnualHoliday, OneOffHoliday;`

2.2.7 HolidayReference

A placeholder for a `Holiday`.

See Also

`Holiday;`

2.2.8 NonBusinessDayRule

The interface that defines a non-business-day rule.

Methods

The `NonBusinessDayRule` interface defines the following methods:

```
public String getName();  
Returns the name of the rule.
```

```
public boolean isNonBusinessDay(SerialDate date); Returns  
true if the specified date is a non-business-day, and false oth-  
erwise.
```

```
public SerialDate nextNonBusinessDay(SerialDate date);  
Returns the next non-business-day according to this rule, or null  
if there are no further non-business-days for the rule.
```

See Also

`Holiday`;

2.2.9 NonBusinessDaySchedule

The interface that defines a schedule of non-business days.

See Also

`NonBusinessDayRule`;

2.2.10 OneOffHoliday

A rule that defines a single date as a holiday. Useful to cover holidays that don't follow a regular pattern.

See Also

`Holiday`;

2.2.11 Weekend

Represents the days in a weekend (usually `SATURDAY` and `SUNDAY`).

See Also

`Holiday`;

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